

CollectionsLib Unit Testing using NUnit

Introduction

This experiment explains how to write automated unit test cases for the CollectionsLib project using the NUnit framework. The test cases validate different collection scenarios such as null values, employee existence, uniqueness of records, and comparison of two employee collections.

Objectives

- Learn automated unit testing using NUnit.
- Understand NUnit custom attributes.
- Learn how to use CollectionAssert.
- Verify collection data using Classic Model and Constraint Model assertions.

Implementation

Step 1: Create a Class Library project named CollectionsLib.Tests in the same solution.

Step 2: Rename the default class to EmployeeTests.cs.

Step 3: Add the reference of CollectionsLib to the test project.

Step 4: Install NUnit and NUnit3TestAdapter using NuGet Package Manager.

Step 5: Create a test method to verify that the employee collection does not contain any null values.

Step 6: Create another test method to verify that an employee with ID 100 exists in the collection.

Step 7: Override Equals() and GetHashCode() methods in the Employee class.

Step 8: Write a test case to verify that all employees in the collection are unique.

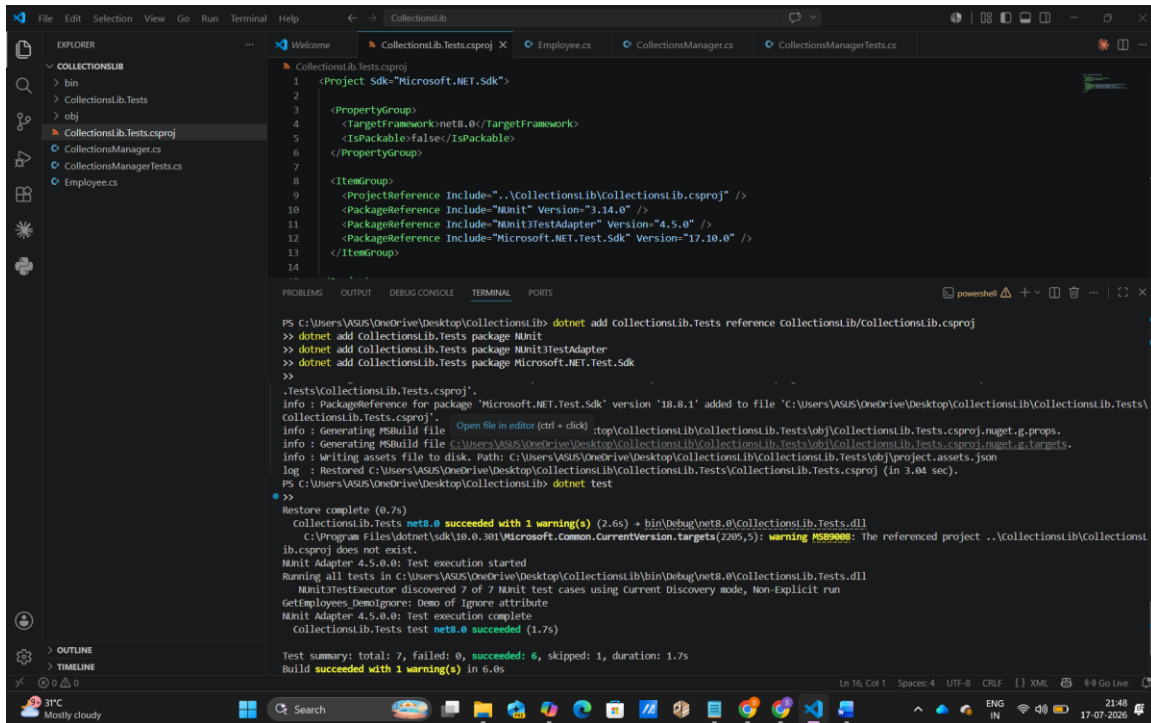
Step 9: Write a test case to compare GetEmployees() and GetEmployeesWhoJoinedInPreviousYears() using CollectionAssert and Assert.That().

Step 10: Build the solution and run all test cases.

Step 11: Modify the source code intentionally to break a test and rerun the tests to observe the failure.

Execution

Paste the screenshot of the NUnit Test Explorer or terminal output here.



```
1 <Project Sdk="Microsoft.NET.Sdk">
2
3   <PropertyGroup>
4     <TargetFramework>net8.0</TargetFramework>
5     <IsPackable>false</IsPackable>
6   </PropertyGroup>
7
8   <ItemGroup>
9     <ProjectReference Include="..\CollectionsLib\CollectionsLib.csproj" />
10    <PackageReference Include="NUnit" Version="3.14.0" />
11    <PackageReference Include="NUnit3TestAdapter" Version="4.5.0" />
12    <PackageReference Include="Microsoft.NET.Test.Sdk" Version="17.10.0" />
13  </ItemGroup>
14</Project>
```

```
PS C:\Users\ASUS\OneDrive\Desktop\CollectionsLib> dotnet add CollectionsLib.Tests reference CollectionsLib\CollectionsLib.csproj
>> dotnet add CollectionsLib.Tests package NUnit
>> dotnet add CollectionsLib.Tests package NUnit3TestAdapter
>> dotnet add CollectionsLib.Tests package Microsoft.NET.Test.Sdk
>>
info : PackageReference for package 'Microsoft.NET.Test.Sdk' version '18.0.1' added to file 'C:\Users\ASUS\OneDrive\Desktop\CollectionsLib\CollectionsLib.Tests\
CollectionsLib.Tests.csproj'.
info : Generating MSBuild file 'C:\Users\ASUS\OneDrive\Desktop\CollectionsLib\CollectionsLib.Tests\obj\CollectionsLib.Tests.csproj.nuget.g.props'.
info : Writing assets file to disk. Path: C:\Users\ASUS\OneDrive\Desktop\CollectionsLib\CollectionsLib.Tests\obj\CollectionsLib.Tests.csproj.nuget.g.targets.
log  : Restored C:\Users\ASUS\OneDrive\Desktop\CollectionsLib\CollectionsLib.Tests\CollectionsLib.Tests.csproj (in 3.04 sec).
PS C:\Users\ASUS\OneDrive\Desktop\CollectionsLib> dotnet test
>>
Restore complete (0.7s)
CollectionsLib.Tests net8.0 succeeded with 1 warning(s) (2.6s) → bin\Debug\net8.0\CollectionsLib.Tests.dll
C:\Program Files\dotnet\sdks\10.0.301\Microsoft.Common.CurrentVersion.targets(2205,5): warning MSB9000: The referenced project ..\CollectionsLib\Collections
Lib.csproj does not exist.
NUnit Adapter 4.5.0.0: Test execution started
Running all tests in C:\Users\ASUS\OneDrive\Desktop\CollectionsLib\bin\Debug\net8.0\CollectionsLib.Tests.dll
NUnit3TestExecutor discovered 7 of 7 NUnit test cases using Current Discovery mode, Non-Explicit run
GetEmployees Demo: Demo of Ignore attribute
NUnit Adapter 4.5.0.0: Test execution complete
CollectionsLib.Tests test net8.0 succeeded (1.7s)

Test summary: total: 7, failed: 0, succeeded: 6, skipped: 1, duration: 1.7s
Build succeeded with 1 warning(s) in 6.0s
```

Expected Result

The employee collection should not contain null values, employee ID 100 should exist, all employee records should be unique, and both collections should satisfy the expected comparison conditions.

Conclusion

This experiment improved the understanding of NUnit collection testing. Different assertion techniques such as CollectionAssert and Assert.That() were used to validate collection data effectively.